

AgentCore Platform — Complete Database Architecture Guide

Version:

Status:

Audience:

What Is This Database For?

enterprise-grade platform for building, deploying, and managing AI agents

operating system layer

How It Is Structured

nine tiers



Tier 1: Organizational Hierarchy

who owns what

Organization

root tenant

-
- tier
- status
-

Example:

Department

subdivision within an organization

-
- parent department
-
-
-

Example:

Project

workspace for organizing agents and resources

-
- sub-projects

-
-

Example:

Environment

deployment stages

-
- production
- require approval

Example:

Tier 2: Identity and Authentication

who users are

where they belong

User

person

-
- Azure Entra ID
-
-
-

Example: john.doe@motherson.com

User Organization Membership

user to an organization

-
-
-

Example:

User Department Membership

user to a department

-
-

Example:

User Project Membership

user to a project

-

-

Example:

Tier 3: Roles, Permissions, and Access Control

who can do what

Role

named set of capabilities

- scope
- hierarchy level
- parent role
-

Example:

Permission

specific action

-
-

Example:

`model.onboard`

`agent.create`

`agent.deploy`

`kb.upload`

`audit.view`

Role Permission

which permissions belong to which roles

-
- conditions

Example:

`agent.create`

`audit.view`

`agent.deploy`

Resource Permission

fine-grained, per-resource access

-
-
-

Example:

Permission Cache

performance optimization

-
-
-

User Role Assignment

actual assignment

-
-
-

Example:

Entra Group Role Map

Azure Entra ID groups to platform roles

-
-

Example:

Tier 4: Agent and Workflow Management

core of the platform

Agent

central entity

-
- lifecycle status
- visibility controls
-
- component
-

Example:

Agent Definition

immutable snapshot

-
-
-

Why immutable?

Agent Version

numbered release

-
-
- current

-
-

Example:

Agent Deployment

where each agent version is running

-
-
- health monitoring
- rollback

Example:

Agent Bundle

frozen snapshot

-
-

Example:

Agent State Transition

audit trail

-
-

Example:

Agent Registry

marketplace listing

-
-
-

public

featured

Example:

Agent Execution

every time an agent runs

-
-
-
-
-

performance metrics

cost

Example:

Agent Execution Step

single execution into individual steps

-
-

Example:

Tier 5: Legacy Flow and Message Support

backward compatibility

Flow

original agent definition format

-
-
-

Session

conversation

-
-
-
-

Example:

Message

individual message

-
-
-

Transaction

execution state

Tier 6: Governance and Approval

nothing reaches production without proper authorization

Approval Request

formal request

-
-
-

Example:

Approval Step

individual step

-
-
-

Example:

Resource Visibility

who can see a resource

-
-

Example:

Resource Access Request

request access

-

-

Example:

Tier 1 : Knowledge Management

Retrieval-Augmented Generation

Embedding Version

which AI model was used to create vector embeddings

-
-

Why this matters:

Vector DB Connection

connection details for external vector databases

-
-
-
-

default

Example:

Knowledge Base

curated collection of documents

-
-
-
-

classification

Example:

Knowledge Base Access

who can access a specific knowledge base

-
-
-
-

Document

every file uploaded

-
- ingestion pipeline
-
- vector database sync status

Example:

Knowledge Item

individual chunk

-
-
- vector embedding
-
-

Example:

Vector DB Sync Log

every synchronization

-
-
-

Knowledge Retrieval Log

every time an agent searches a knowledge base

-
-
-

Example:

Tier 8: Infrastructure and Configuration

external services and tools

Guardrail

safety and compliance rules

-
-
-

Example:

MCP Server

external tool server

-
-
-

Example:

Model Catalog

registry of AI models

-
-
-

Example:

Model Version

specific versions

-
- default

Model Onboarding Request

formal request

-
-
-

Example:

API Key

programmatic access

-
- specific agents specific endpoints
- rate limiting
-
-

Example:

File

file metadata

-
-

Variable

user-defined configuration variables

-
-

Publish Record

external publishing

-
-

Tier 9: Audit and Compliance

complete traceability

Audit Log

comprehensive, append-only record

- action
- entity
- actor
- before and after snapshots
- result
- request context

Why this matters:

partitioning by month

Automatic triggers

How Multi-Tenancy, Security, and Scalability Work

Multi-Tenancy

```
org_id
```

Row-Level Security

-
-
-

Security

Layer	Mechanism
Tenant isolation	
Access control	
Fine-grained access	
Authentication	
Data encryption	
API security	
Audit	
Soft deletes	

Scalability

Concern	Approach
Query performance	
Permission checks	
Audit volume	
Vector search	
Full-text search	
Soft deletes	

Summary

Dimension	Capability
Purpose	
Tables	
Tenancy	
Access Control	
Agent Management	
Knowledge	
Governance	
Infrastructure	
Audit	
Integration	
Security	

This database schema provides the foundation for a platform where AI agents can be built, governed, and operated with the same rigor expected of any enterprise-critical system.